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Dr. Tianfei Xue
GEOMAR Helmholtz Centre for Ocean Research Kiel
Biogeochemical Modelling
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Application as Postdoctoral Researcher (m/f/d) in Ocean-Biogeochemical Modelling – ZOOM-IN

Dear Dr. Xue,

I have spent over three years developing biogeochemical models for plankton dynamics within the GOTM-FABM/ERGOM framework, implementing biological parameterizations in Fortran and running climate scenario simulations. The ZOOM-IN project's goal of implementing zooplankton diel vertical migration (DVM) parameterizations within a regional ocean circulation–biogeochemical model is a natural extension of this work, and I am highly motivated to contribute my expertise to your research team.

During my postdoctoral research at the University of Hamburg, I developed the “Cyanobacteria Life Cycle” (CLC) model within the ERGOM/GOTM-FABM framework. A key achievement was taking a published theoretical framework (Beckmann 2019) and implementing its trait-diffusion scheme within an existing ocean modelling infrastructure in Fortran. In parallel, I built a Lagrangian individual-based model (IBM) for phytoplankton dynamics incorporating stochastic processes, individual variability, and trait evolution. This experience with both Eulerian and Lagrangian approaches within a coupled physical-biogeochemical model framework prepares me directly for adapting DVM parameterizations into the CROCO-BioEBUS system.

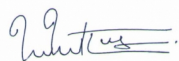
My research heavily involved conducting hindcast simulations using ERA5 data and forward projections under warming scenarios to assess ecosystem responses. I regularly executed model ensembles on HPC clusters (SLURM, JSC Jülich) and analyzed the resulting large NetCDF datasets using Python (NumPy, xarray). Furthermore, I bring dual competence in classical Fortran modelling and modern computational techniques, such as using Python/JAX for GPU/TPU scaling of models.

Your recent work on the interplay between bottom-up and top-down controls on phytoplankton (e.g., Xue et al., *Biogeosciences*, 2024), particularly how changes in the physical environment impact zooplankton grazing pressure, strongly resonates with my own findings on phytoplankton responses to environmental drivers. I am eager to extend this perspective to the Peruvian upwelling system, investigating how zooplankton DVM connects surface productivity with nitrogen loss in the oxygen minimum zone.

I am currently preparing a manuscript on my IBM trait-diffusion modelling results in collaboration with Prof. Dr. Elisa Schaum (University of Hamburg) and Prof. Dr. Kai Wirtz (Helmholtz-Zentrum Hereon). I also contributed to teaching Marine Ecosystem Modelling at BSc and MSc level at the University of Hamburg.

I am available to start from July 1, 2026. I have enclosed my CV, including contact details for two references, along with my certificates. I look forward to the opportunity to discuss how my background can support the ZOOM-IN project.

Kind regards,



Dr. Thejus Mahajan

Enclosures: Curriculum Vitae (with references), Certificates